Experiment 10

BI Deployment, Monitoring and Backup & Restore using Microsoft Power BI

Aim: To understand and design Business Intelligence deployment, monitoring, backup, and recovery strategies using Microsoft Power BI for ensuring availability, reliability, and continuity of BI solutions.

OBJECTIVES:

1. To identify and plan the deployment of BI assets using Power BI.
2. To design workspace structures and manage user access roles.
3. To develop monitoring strategies for dataset refresh, dashboard usage, report performance and user activity.
4. To plan backup and restore procedures for BI components.
5. To design disaster recovery strategies to ensure business continuity.

TOOLS / SOFTWARE REQUIRED:

1. **Operating System:** Windows 11.
2. **Software:** Microsoft Power BI Desktop.
3. **Dataset**: Global Superstore dataset in CSV or Excel format.
4. Power Query Editor
5. DAX (Data Analysis Expressions)
6. **System Requirements:** Minimum 4 GB RAM.

DATASET USED: Global Superstore Dataset.xlsx.

THEORY:

Business Intelligence (BI) deployment involves making reports, dashboards, datasets and analytical solutions available to authorized users across an organization. It ensures that business information can be accessed efficiently for better decision-making.

Monitoring is an important part of BI management. It helps track dataset refresh status, dashboard usage statistics, report performance and user activity. Regular monitoring helps identify issues and maintain the reliability of BI solutions.

Backup and restore strategies protect BI assets from accidental deletion, data corruption and system failures. Regular backups of datasets, reports and workspace content help recover important information whenever required.

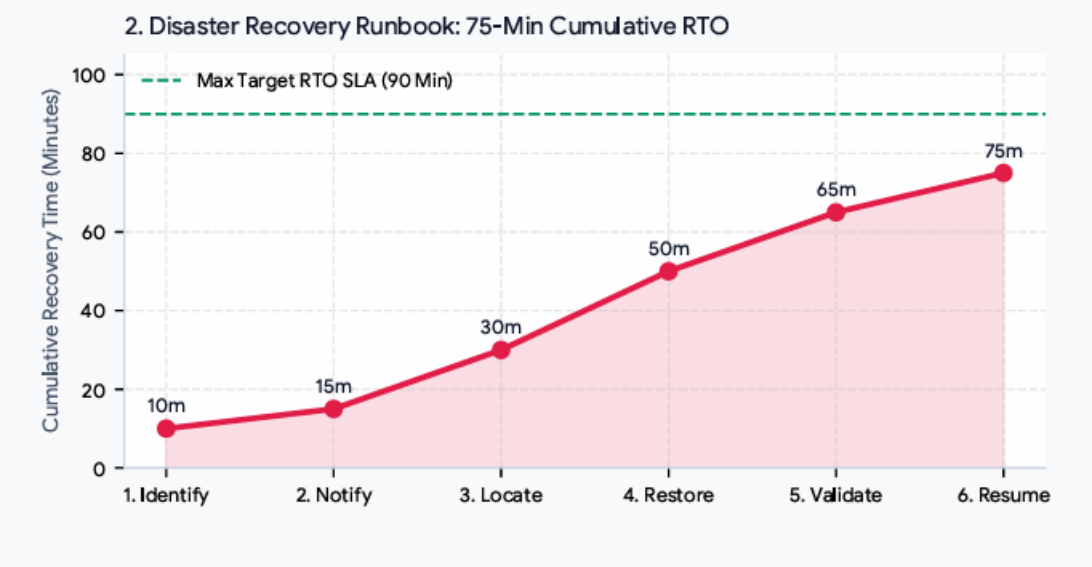
Disaster recovery planning defines the procedures for restoring BI services during unexpected incidents. It helps minimize disruptions, maintain business continuity and ensure the availability of critical business information.

Microsoft Power BI provides tools through Power BI Desktop and Power BI Service to develop, publish, manage and monitor BI solutions effectively.

PROCEDURE:

Part A: BI Deployment Planning

1. Identified the BI assets created in previous experiments, including reports, dashboards, datasets and workspaces.
2. Listed the reports, dashboards, datasets and workspaces that require deployment.
3. Prepared a BI Deployment Plan to organize the deployment process.
4. Identified the intended users and departments, including Executive Management, Sales, Finance and Regional Management.
5. Defined deployment objectives and expected outcomes to ensure effective availability and accessibility of BI solutions.



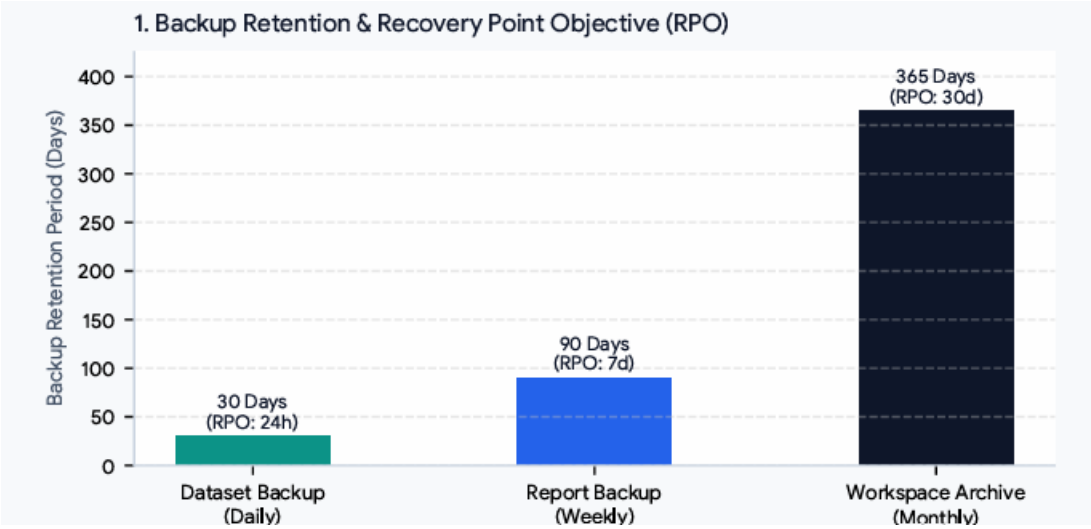
Part B: Workspace and Access Planning

6. Designed a centralized Power BI Workspace named Global Superstore BI Workspace to manage reports, datasets and dashboards.

7. Identified four workspace roles: Administrator, Member, Contributor and Viewer.

8. Assigned responsibilities and access permissions to each role according to their functions.

9. Documented workspace governance guidelines, including access control, permission reviews, naming conventions, data security and regular backups.



Part C: Monitoring Strategy Design

10. Identified the monitoring requirements, including Dataset Refresh Status, Dashboard Usage Statistics, Report Performance and User Activity Monitoring.

11. Designed a monitoring strategy by defining monitoring metrics, frequency and responsible personnel for each monitoring activity. Established daily and weekly monitoring schedules to maintain BI system performance and reliability.

Part D: Backup and Restore Planning

12. Identified the BI components requiring backup, including datasets, Power BI reports and workspace content.

13. Developed a backup schedule consisting of daily dataset backups, weekly report backups and monthly workspace archives.

14. Designed a restore procedure to recover BI solutions in case of accidental deletion or system failure by locating the latest valid backup, restoring the required components and validating their functionality.

2. Power BI Workspace Role Capability Matrix

Capability / Action	Admin	Member	Contrib.	Viewer
View & Interact with Reports	Yes	Yes	Yes	Yes
Create / Edit Reports & Datasets	Yes	Yes	Yes	No
Publish & Update Workspace App	Yes	Yes	No	No
Add / Remove Workspace Users	Yes	Yes	No	No
Delete Workspace & Configure Settings	Yes	No	No	No

Result

Successfully designed a BI deployment, workspace and access management, monitoring, backup and restore, and disaster recovery strategy using Microsoft Power BI for the Global Superstore scenario. The report provides a structured approach to maintaining BI solution availability, reliability and business continuity.

LEARNING OUTCOMES:

- Understood the deployment requirements of Business Intelligence solutions.
- Learned to design Power BI workspace structures and manage user access roles.
- Understood monitoring strategies for dataset refresh, dashboard usage, report performance and user activity.
- Learned to develop backup and restore plans for BI components.
- Gained knowledge of disaster recovery planning to ensure business continuity.